DNS Address Book Activity

We just completed an activity that translated between human-readable names (people) and computer-based IP addresses (numbers). Students were given a slip of paper containing an IP address. Their goal was to create and maintain a written list of every one of their classmates' IP addresses. The only rules were:

- Students may walk around the room.
- Students may share information with classmates.
- Students may talk or share information with only one classmate at a time.
- Students must write the IP address list on a piece of paper.

But as the students worked, I circulated quietly through the room and exchanged their IP address slip for another. My goal was to make it increasingly difficult for them to maintain an accurate list.

Discussion Questions

1. Why do you think I was switching your IP addresses?

2. What made the exercise difficult: the number of classmates, the changing IP addresses, or the fact that information spread person-to-person? Why?

3. The activity can become annoying quickly. If we had increased to 30 million people, what new problems would appear?

4. What could go wrong if someone gave you the wrong IP address?

5. IP addresses can change; is there a better system to know everyone else's IP address?

IP Address Log

No.	Name	IP Address	No.	Name	IP Address
1.			2.		
3.			4.		
5.			6.		
7.			8.		
9.			10.		
11.			12.		
13.			14.		
15.			16.		
17.			18.		
19.			20.		
21.			22.		
23.			24.		
25.			26.		
27.			28.		
29.			30.		